Bihar Engineering University, Patna
VIII[th] Semester Examination - 2023

Course: B.Tech. | *Time: 03 Hours*
Code: 101818 | *Subject: Railway Engineering* | *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Rails is fixed to steel sleeper by
(i) welding (ii) bolts and nuts
(iii) riveting (iv) using keys between lugs or loose joints

(b) The rail is designated by its
(i) length (ii) weight
(iii) cross- section (iv) weight per unit length

(c) Sleeper density in India is normally kept as
(i) M+2 to M+7 (ii) M to M+2
(iii) M+5 to M+10 (iv) M Where M is the rail length in meters

(d) The ballast below the sleeper is called
(i) ballast cushion (ii) crib ballast
(iii) shoulder ballast (iv) heavy ballast

(e) Number of cotters used in CST-9 sleepers
(i) 2 (ii) 3
(iii) 4 (iv) 5

(f) Creep is the
(i) longitudinal movement of rail (ii) lateral movement of rail
(iii) vertical movement of rail (iv) difference in level of two rails

(g) Which one of the following measures the stiffness of track?
(i) Tractive effort (ii) Tractive resistance
(iii) Track modulus (iv) Load capacity

(h) The maximum degree of curvature for Metre Gauge is limited to
(i) 10^0 (ii) 16^0
(iii) 30^0 (iv) 40^0

(i) The shape of transition curve used by Indian railway is
(i) cubic parabola (ii) spiral
(iii) sine curve (iv) lemniscates of Bernoulli

(j) Which one of the following characteristics of ballast makes it unsuitable for use?
(i) High resilience (ii) High stability
(iii) High water absorption (iv) High modulus

Q.2 (a) Explain different types of gauges used in India. **[7]**
(b) What is creep of the rail? Explain the all theories of creep of the rails. **[7]**

Q.3 (a) What is the ballast in permanent way? Mention the functions of ballast and state the requirement of good ballast material. **[7]**
(b) Explain the basic principles of railway track alignment and the factors considered in selecting the alignment for new railway line. **[7]**

© Apna BEU
HIGHLY CONFIDENTIAL
Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.4 (a) The wheel base of a vehicle moving on a BG track is 6 m the diameter of the wheels is 1524 mm and the flanges project 32 mm below the top of the rail. Determine the extra width of the gauge required if the radius of the curve is 168 m. Also indicate the extra width of gauge actually provided as per Indian Railway Standards [7]

(b) What are the design details in a diamond crossing? [7]

Q.5 (a) Enumerate the various types of track junctions in use. Discuss the suitability of each under different situations. [7]

(b) Calculate the cant deficiency and permissible speed for a 4° curve of BG track. Also find the equilibrium cant and length of transition curve. [7]

Q.6 (a) Calculate the maximum permissible train load that can be pulled by a locomotive having four pairs of driving wheels carrying an axle load of 24 tonnes each. The train has to run speed 80 km/hr on a straight level track(BG). [7]

(b) Discuss in brief the various tractive resistance in railways. [7]

Q.7 (a) Explain with neat sketches, how surface and sub-surface water can be removed from a railway track. [7]

(b) Explain the process to manage the impact of thermal expansion and contraction on railway tracks, especially in region with extreme temperature variations. [7]

Q.8 What is super elevation? What are the objects of providing super elevation on a curve of rail track? [14]

Q.9 Write short notes on any four of the following: [3½ x 4=14]

(a) Track drainage
(b) Sleeper density
(c) Rail failures
(d) Types of rail joints
(e) Tractive effort and hauling capacity
(f) Length of transition curve.

© Apna BEU
CONFIDENTIAL DOCUMENT EDC-45 Dt. 25-08-2023

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play